

Lane Drew

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PROFESSIONAL APPOINTMENTS

University of Colorado Anschutz

Postdoctoral Fellow

Department of Biostatistics and Informatics

December 2025 – Current

EDUCATION

Colorado State University

PhD, Statistics, Advisor: Dr. Andee Kaplan

Fort Collins, CO

August 2019 – December 2025

Colorado State University

MS, Statistics, Advisor: Dr. Andee Kaplan

Fort Collins, CO

August 2019 – May 2023

University of New Mexico

BS, Statistics & Psychology, summa cum laude

Albuquerque, NM

August 2008 – May 2012

RESEARCH INTERESTS

Bayesian Statistics, Entity Resolution and Record Linkage, Statistical Computing, Spatial Statistics, Point Processes, Machine Learning, Longitudinal MNAR

HONORS AND AWARDS

CSU Department of Statistics Boes Teaching Award

May 2025

CSU College of Natural Sciences Thomas A. Jones Graduate Fellowship

August 2024

EnviBayes Workshop on Complex Environmental Data Registration Award

September 2023

CSU Department of Statistics James R. zumBrunnen Statistical Collaboration Award

May 2023

CSU Department of Statistics Department Distinguished Service Award

May 2023

JOURNAL PUBLICATIONS

Refereed

- **Drew, L.**, Kaplan, A., and Breckheimer, I. “A Bayesian Record Linkage Approach to Applications in Tree Demography Using Overlapping LiDAR Scans”. *Annals of Applied Statistics* 19.3 (2025), pp. 2027–2052.

Preprints

- **Drew, L.**, Kaplan, A. “ldmppr: Location-Dependent Marked Point Processes in R”. *arXiv*, arXiv:2605.19100 (2026).

TALKS, SOFTWARE, AND POSTER PRESENTATIONS

Contributed Talks

- “A Bayesian Record Linkage Approach to Tree Demography Using Overlapping LiDAR Scans”. *WNAR*. Student Paper Presentation. International Biometric Society. Fort Collins, CO, June 2024.
- “Bayesian Spatial Record Linkage as Applied to Tree Demography Using Overlapping LiDAR Scans”. *Spatial Statistics*. Topic Contributed Talk. Elsevier. Boulder, CO, July 2023.

Software

- **Drew, L.**, Kaplan, A. *ldmppr: Estimate and Simulate from Location Dependent Marked Point Processes*. R package version 1.0.4. 2024.

Posters

- “A Bayesian Record Linkage Approach to Tree Demography Using Overlapping LiDAR Scans”. *ENVR Workshop: Spatial Data Science for the Environment*. Contributed Poster. ASA. Boulder, CO, October 2024.
- “A Bayesian Record Linkage Approach to Tree Demography Using Overlapping LiDAR Scans”. *EnviBayes Workshop on Complex Environmental Data*. Contributed Poster. ISBA. Fort Collins, CO, September 2023.
- “A Bayesian Record Linkage Approach to Tree Demography Using Overlapping LiDAR Scans”. *Conference on Data Analysis*. Contributed Poster. LANL. Santa Fe, NM, March 2023.

RESEARCH EXPERIENCE

Graduate Research Assistant

Summer 2021/2022/2023/2025

Colorado State University

Fort Collins, CO

- Developed and implemented frameworks for Bayesian entity resolution and deduplication models using remote sensing data alongside Dr. Andee Kaplan and Dr. Ian Breckheimer (Rocky Mountain Biological Laboratory) for applications in tree demography.

CONSULTING EXPERIENCE

Prison Agriculture Lab

Fall 2022/2025

Colorado State University

Fort Collins, CO

- Worked alongside the lab directors, Dr. Carrie Chennault and Dr. Josh Shicca, to develop a comprehensive national dataset by integrating data from multiple sources using advanced record linkage techniques.
- Constructed predictive models to identify key drivers of agricultural programs within the US prison system.

TEACHING EXPERIENCE

Graduate Teaching Assistant

August 2019 – Present

Colorado State University

Fort Collins, CO

- Instructor of record for STAT 100 Statistical Literacy (SP 23, FA 24), STAT 158 Introduction to R Programming (SU 23, SU 24, SP 25), STAT 201 General Statistics (SU 20), STAT 204 Statistics with Business Applications (SU 20), and STAT 301 Introduction to Applied Statistical Methods (FA 21, SP 22, FA 22, SP 24).
- Served as a grader for upper division undergraduate and graduate level statistics courses including STAT: 315 (FA 19), 421 (SP 20), 430 (SP 20), 440 (SP 20), 600 (SP 24), STAR: 511 (FA 21/22), 513 (SP 22), STAA: 556 (SU 24).
- Tutored students in introductory-level statistics courses in the Statistics Success Center at Colorado State University including STAT: 100, 158, 201, 204, 301, 307, 315.

Coding & Cookies Instructor

March 2022 – November 2024

Colorado State University

Fort Collins, CO

- Workshop facilitator for Version Control Using git (SP 24, FA 24), Data Visualization Using ggplot (SP 22), and R Basics (FA 22, SP 23, FA 23).

PROFESSIONAL EXPERIENCE

Restaurant Manager

July 2016 – August 2019

The Shed

Santa Fe, NM

- Managed a team of approximately 50 staff across hosting, service, bar, and kitchen areas, ensuring efficient operations and high-quality customer service in a high volume business.
- Developed employee training programs that enhanced performance and fostered a collaborative work environment.

Sampler & Lab Technician

September 2009 – August 2019

Indepth Water Testing

Santa Fe, NM

- Collected and analyzed samples, with strict adherence to quality control standards and regulatory compliance.
- Conducted microbiological and chemical analyses, compiling detailed reports to communicate findings to clients.
- Collaborated with communities on sampling protocols, ensuring reliability of environmental assessments.

GRADUATE COURSEWORK

STAT: 520 Intro to Probability Theory, 525 Analysis of Time Series I, 530 Mathematical Statistics, 540 Data Analysis and Regression, 555/556 Statistical Consulting Sequence, 620 Introduction to Measure Theoretic Probability, 630 Advanced Statistical Data Analysis, 640 Design and Linear Modeling I, 670 Bayesian Statistics, 720 Probability Theory, 730 Advanced Theory of Statistics I, 740 Advanced Statistical Methods (Point Processes)

TECHNICAL SKILLS

Languages: R, STAN, Bash, L^AT_EX, Markdown

Developer Tools: Git, RStudio, Positron

Libraries: Rcpp/RcppArmadillo, rstan, ggplot2, tidyverse, tidymodels, devtools

AFFILIATIONS

American Statistical Association, International Society of Bayesian Analysis, Institute of Mathematical Statistics